

Circuits: How Does a Torch Work?

Grade 1 Science Lesson

1. Why Do We Need Torches?

Imagine it's nighttime and suddenly all the lights go off! The room becomes very dark and you can't see your toys or books. What do we reach for? **A torch!**

In India, every family keeps a torch ready for power cuts to find their way safely. Watchmen and street vendors also use them to work at night.

2. The 4 Main Parts of a Torch

The Battery

This stores all the **energy** inside. Think of it like a tiffin box that stores yummy food for later!

The Bulb

This is the part that **glows bright**. Inside is a tiny wire called a filament that gets hot and makes light.

The Switch

You push this to turn the torch **ON or OFF**. It works like a door for electricity.

The Body

The outer case that holds everything together **safely**.

3. How Electricity Moves

"Electricity moves in a circle, like children running around a track!"

- **Switch ON:** The path is complete! Electricity flows and the bulb glows like a little sun.
- **Switch OFF:** The path is broken. Electricity cannot flow, so the bulb stays dark.

What if things change?

- **No Battery:** No energy, so no light!
- **Battery Backwards:** If put in the wrong way, the electricity cannot flow.

Activity: Let's Draw Our Torch!

Follow these steps to draw your own torch:

1. Draw a **long rectangle** for the body (Color it Saffron-Orange).
2. Add a **small circle** at the top for the bulb (Color the light Sunshine Yellow).
3. Draw a **tiny rectangle** on the side for the switch (Color it Peacock Blue).
4. Label the parts: **Battery, Bulb, and Switch.**

4. Fun Science Experiment

Color Magic!

Try holding **colored paper** in front of your torch light. Red paper makes red light! This happens because white torch light has all colors hiding inside, and the paper only lets one color through.

Source: Torch Power Circuits for 1st Grade Instructional Material

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